Senior AI Engineer – Langfuse –EKS – SustainRecruit- Hollard? Ms Ffion Bushell – 12/10/2025 just expired when apply. Asked Petra to pass on, but no resp.
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Sr. AI Engineer / GenAI | Automation Engineer/ Sr. DevOps /Data Engineer /Full Stack Developer (Python/Java/.Net/Mulesoft /AI/ML/DL/BI/SharePoint/Dynamics 365)-Japan (7.5yrs) & Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Sr. AI Engineer |Automation Engineer | Software Engineer /Full Stack Developer/Sr. DevOps /Data Engineer
Responsibilities:

- **Strategy & Platform Architecture (AI Vision & Leadership)**
- Spearheaded the design, build, and scaling of Generative AI and agentic LLM-powered solutions across multiple cloud platforms, delivering clinical and operational intelligence by translating complex requirements into usable, user-facing AI/ML products that drive transformation in the healthcare domain.
- Applied deep technical AI/ML expertise with a product-focused mindset across the full AI development lifecycle, from prototyping and prompt engineering through to secure, production-grade deployment of intelligent solutions.
- Drove the full lifecycle of AI solutions, from ideation and opportunity identification with key stakeholders to the design and delivery of scalable prototypes. Conducted comprehensive experiments to evaluate model performance, iterating on design to optimize outcomes and ensure compliance with ethical AI standards.
- Consistently delivered high-quality AI solutions within agile sprint cycles, using data-driven metrics like MTTR and FPR to evaluate performance and iterate on improvements, which allowed for effective communication of complex AI logic to non-technical stakeholders via dashboards and chat interfaces.
- **Generative AI & RAG (LLM Engineering & Agentic Workflows)**
- Engineered & optimized high-impact LLM-driven solutions for clinical intelligence, including multi-document summarization, structured data extraction, and autonomous workflow automation. Architected and deployed low-latency, production-grade Retrieval-Augmented Generation (RAG) pipelines, leveraging specialized tooling (e.g., LangChain, LlamaIndex, vector databases) with optimized chunking strategies & asynchronous endpoints (FastAPI) for real-time performance monitoring (Langfuse). Example: response = RAGChain.run({"question": clinical_query, "docs": retrieved_context})
- Architected and scaled a production-grade RAG framework featuring advanced context management and layered prompt engineering, achieving high-fidelity clinical data retrieval. Integrated an LLM-as-an-Optimizer module to automatically test and refactor retrieval strategies based on real-time performance metrics, dramatically improving model safety and response quality. E.g., llm_output = bedrock_client.invoke_model(modelId='anthropic.claude-v3', or 'amazon.titan-text-express-v1' body={"prompt": "Refactor this code to optimize performance: {code_snippet}
- Developed and implemented an advanced Agentic Graph RAG framework, leveraging advanced Graph RAG methodologies to construct Knowledge Graphs from diverse unstructured data sources. This architecture enabled relationship-aware queries and complex multi-hop reasoning, achieving a 25% improvement in accuracy and significantly reducing hallucination rates for knowledge-intensive clinical platforms.
- Developed and orchestrated sophisticated autonomous AI agents using frameworks like LangGraph, AutoGen, and Semantic Kernel, and leveraging inter-agent communication protocols and tools like n8n to integrate with external APIs, driving operational efficiency and intelligent decision-making.
  E.g, response = await kernel.invoke_async("email_plugin.summarize", text=email_body).
- Engineered and optimized end-to-end RAG pipelines, including advanced Graph RAG and intelligent chunking with LangChain, to enhance contextual understanding and answer complex semantic queries, boosting performance and reducing API calls by over 20%.
  E.g. 1: text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200).
  E.g. 2: async def query_llm_with_rag(index: VectorStoreIndex, question: str) -> Dict[str, Any]:
- Architected and deployed enterprise-grade Generative AI solutions using Amazon Bedrock, leveraging a suite of pre-trained foundation models including Anthropic's Claude 4 and Amazon Titan. Applied prompt engineering and LLM-based RAG to build a context-aware customer support chatbot, reducing response time by 30%.
- Led end-to-end GenAI solution development from secure audio ingestion, automated transcription, LLM orchestration, RAG pipeline integration, to intelligent summarization—ensuring seamless data flow across multi-cloud.
- **Machine Learning & Model Development (Core ML Expertise)**
- Developed and trained robust machine learning models—including those for prediction, anomaly detection, and regression—for high-impact clinical applications, leveraging deep feature engineering expertise across structured and unstructured datasets. E.g.1: clf.fit(patient_vitals_features, outcome_labels)
  E.g.2: model.fit(NLP_features_matrix, length_of_stay_days)

- Developed, trained, and deployed a suite of machine learning and deep learning models on Amazon SageMaker using PyTorch, TensorFlow, and Hugging Face Transformers, optimizing data pipelines and automating model retraining to improve predictive accuracy and deliver NLP solutions.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive Next Best Action (NBA) recommendations and pattern discovery.

- **MLOps, Deployment & Infrastructure (Scaling & Reliability)**
- Integrated ML/AI services into core clinical applications using modern cloud and containerized deployment methods (e.g., Docker, AWS ECS/EKS/Lambda), ensuring high availability and robust scalability across the production environment.
- Architected & implemented production-grade monitoring & evaluation (M&E) frameworks for high-volume LLMs & predictive ML models, leveraging a mix of specialized observability tooling (e.g., Langfuse) & cloud-native services (e.g., Amazon CloudWatch) to track performance drift & enforce data quality and security guardrails across scalable deployment pipelines. E.g., Langfuse.trace_guardrails(response, checks=["PII_redaction", "hallucination_score"])
- Pioneered the use of LLMs and agentic workflows to automate and refactor code within scalable CI/CD pipelines, reducing deployment time by 25%. Engineered and deployed mission-critical Python applications and GenAI platforms on a multi-cloud architecture (AWS, Azure, GCP), leveraging IaC, MCP+RAG, and n8n orchestration.
- E.g.1: llm_output = gemini.generate_text(f"Refactor this code to optimize performance: {code_snippet}")
- E.g.2: ansible-playbook provision_rhel_app_servers.yml
- E.g.3: requests.post(f"{api_url}/api/v1/deployments", json={'app':'genai-service', 'version':'v1.0.0'}).
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs).
- Implemented cloud-native architectures with AWS ECS/EKS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms streamlined data ingestion within an AWS Redshift-powered lakehouse.
- Designed and developed scalable backend APIs and seamless integrations using Python frameworks (Django, Flask), optimized data pipelines for AI model inference, enabling real-time predictions on large datasets.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Sr. AI/ML Engineer /Automation Engineer/Sr. DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- **Technical Foundations & Modeling (Core ML Expertise)**
- **Built and secured predictive deep learning models** and machine learning APIs using diverse algorithms (CNN, RNN, GAN, reinforcement learning) and applying PyTorch, TensorFlow, Scikit-learn, and Keras to **drive next best action (NBA) recommendations, pattern discovery, and autonomous decision-making.**
- Developed Multi-agent systems (MAS) in Python, demonstrating early expertise in agent-based workflows and automated optimization that enabled systems to autonomously adapt to complex, high-stakes environments.
- Applied advanced computer vision techniques (OpenCV) for anomaly detection and automated document validation, and utilized NLP (NLTK, spaCy) for high-volume text analysis and intelligent summarization.
- Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for machine learning services.
  E.g., requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'}).
- **MLOps, Deployment & Infrastructure (Cloud & Scale)**
- Pioneered the automation of secure, scalable IaC CI/CD pipelines and isolated machine learning platforms, enabling robust multi-cloud deployment (AWS, Azure, GCP) of Python-based server-side applications (Django, Flask).
- Implemented cloud-native architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion within a **data lakehouse architecture leveraging S3 for storage and RedShift for high-performance analytics.**
- Specialized in designing and implementing end-to-end automation solutions using Python and Ansible to streamline secure, low-latency infrastructure provisioning for core production systems.
  E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- Integrated PagerDuty with Amazon DevOps Guru to leverage its AIOps/ML-powered operational insights, translating them into actionable incidents, triggering a PagerDuty incident. Runbook Automation via Bamboo: Reduce support costs by 50% by eliminating toil with PagerDuty. E.g., aws cloudwatch put-metric-data --metric-name CriticalApplicationError --namespace MyAIOpsApplication --value 1 --unit Count --dimensions Environment=Production,Service=AuthService.
- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing and natural language processing. E.g., model.predict(vectorizer.transform(["RBA interest rate hike triggers AUD volatility"]))
- Led end-to-end machine learning solution development from secure audio ingestion, automated transcription, to intelligent summarization—ensuring seamless data flow across multi-cloud.Integrated speech-to-text APIs for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.

- Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, and RHEL9 environments. E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations.E.g. 1: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }E.g. 2: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. ML Engineer/Automation Engineer /Integration Solution Architect /ML/DL/Mulesoft Developer
Responsibilities:

- **Machine Learning Foundations & Early MLOps**
- Pioneered the development of self-learning Multi-agent systems (MAS) in Python to autonomously manage and optimize complex administrative and data flows, establishing early expertise in agent-based control logic.
- Developed computer vision models (using OpenCV and Scikit-image) for automated anomaly detection and the analysis of complex visual data, streamlining data processing and enhancing decision-making accuracy.
- Built robust predictive models and secure APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, regression, reinforcement learning), applying Scikit-learn and Keras for pattern discovery and predictive analytics.
- Developed scalable Python applications leveraging core frameworks like scikit-learn and NLTK for data preprocessing, modeling, and natural language processing (NLP). Engineered intelligent rule-based automation (RPA) workflows powered by these NLP libraries to automate complex data analysis and extraction in research and academic projects.
- **Systems Engineering and Automation**
- Automated the creation of secure and scalable infrastructure for machine learning experimentation using Infrastructure-as-Code (IaC) (Terraform) and automated CI/CD pipelines (e.g., Jenkins) for robust deployment across cloud platforms (AWS, Azure, and GCP).
- Designed and developed scalable backend APIs and integrations using Python frameworks (Django, Flask), optimizing data pipelines for efficient machine learning model training and enabling near real-time predictions on large datasets.
- Implemented cloud-based architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion for scalable systems (EC2, S3, and RDS).
- Developed an intelligent web scraper (Python, requests, BeautifulSoup, pandas) with a simple UI built using Django or Flask to demonstrate robust data acquisition and processing capabilities.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: ML Engineer /Automation Engineer/Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- **Machine Learning & Clinical Application**
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the clinical compensation domain.
- Developed self-learning Multi-agent systems (MAS) in Python to automate and optimize the processing of injury claims, demonstrating early expertise in agent-based control and autonomous adaptation to clinical data.
- Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of clinical compensation claim data, ensuring data integrity and seamless system integration.
- **DevOps, Cloud Infrastructure & Automation**
- Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery and utilized PowerShell scripting to automate system configurations and testing.
- Automated deployment of scalable machine learning applications and services, leveraging AWS services (EC2) to ensure high availability and stability.
- Architected and implemented data solutions on AWS (EC2, RDS, S3), focusing on traditional data warehousing and ETL processes that supported ML model development and analysis.
- Utilized Amazon Redshift for data warehousing and Amazon EMR/Kinesis for the batch and real-time processing of large data volumes, supporting advanced analytics and reporting essential for clinical insights.
- Created and implemented mapping documents that defined how source data fields corresponded to target data fields, including writing code to map the data according to defined rules.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.
- Automated build and deployment processes using Jenkins and shell scripting, managing server configurations with Puppet to ensure reliable delivery of machine learning models for research projects.
- Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java.
- Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- **Security and Compliance**: Implemented security best practices and regularly patched systems and applications.
- **Agile and Data Expertise**: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Sr. Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Sr. Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Sr. Systems Engineer /DWH
Responsibilities:

- Planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.

- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI API , ChatGPT-4o, Multi-GPTs, Auto-Agents, Amazon Bedrock (Claude/Titan), SageMaker & AWS Kiro; Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Sr. AI Engineer – Nuage Sam in charge AI role, also via Rackspace to end client of CBA -8/10/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Sr. AI Engineer /Software Engineer / AI Build | Automation Engineer/ Sr. DevOps /Data Engineer /Full Stack Developer (Python/Go/Java/.Net/Mulesoft /AI/ML/DL/BI/SharePoint/Dynamics 365)-Japan (7.5yrs) & Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Sr. AI Engineer /Software Engineer/ | Automation Engineer | Full Stack Developer/Sr. DevOps /Data Engineer
Responsibilities:

- **Agentic Platform Development & Generative AI**

- Engineered and optimized state-of-the-art pipelines, including advanced and the agentic framework, integrating structured knowledge graphs and reinforcement learning to enhance multi-hop reasoning and achieve a 25% improvement in semantic query accuracy. E.g.1: text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200).
  E.g. 2: async def query_llm_with_rag(index: VectorStoreIndex, question: str) -> Dict[str, Any]:
- Engineered a multi-cloud solution in Python () to instantly query and validate complex aged care funding and compliance documents, resulting in secure integration of compliance scores back into core ledger systems via enterprise-grade Java () microservices. E.g., java_system.send_event(Topic.COMPLIANCE_REVIEWED, genai_results['validation_score'])
- Developed and implemented an advanced Graph-R1 agentic RAG framework, integrating structured knowledge graphs and reinforcement learning to enhance multi-hop reasoning and achieve a 25% improvement in the accuracy of semantic query responses for a knowledge-intensive platform.
- **High-Performance Fraud Engine (Rust & Python)**:
- Designed and deployed a containerized financial fraud detection service, utilizing Rust for the ultra-low-latency transaction analysis and state management kernel, orchestrated via a scalable Python (Flask) API layer, resulting in a 75% reduction in false positives compared to legacy Java systems.
  E.g., rust_core_response = requests.post(F'http://rust-svc/v1/fraud_check', json={'tx_id': id, 'data': payload}).json()
- **GenAI Policy Compliance (Python & Java)**:
- Engineered a multi-cloud RAG solution in Python (Vertex AI/Gemini) to instantly query and validate complex aged care funding and compliance documents, resulting in secure integration of compliance scores back into core ledger systems via enterprise-grade Java (Spring Boot) microservices. E.g., java_system.send_event(Topic.COMPLIANCE_REVIEWED, genai_results['validation_score'])
- **Real-Time Data Pipelines (Go & Python)**:
- Built real-time financial data ingestion and reconciliation pipelines using lightweight Go microservices for high-throughput messaging, feeding clean data directly into feature stores developed in Python for GenAI-driven anomaly detection models.
  E.g., go_consumer.read_msg().publish(python_feature_store, event.ReconciledData)
- Expertly navigated multi-cloud LLM ecosystems (AWS Bedrock, GCP Vertex AI/Gemini, OpenAI API, and Azure OpenAI/ GPT-4) to develop specialized financial management engines for compliance, anomaly detection, and fraud processing.
  E.g. 1: response = vertex_ai_client.predict(model_id='gemini-2.5-flash', **config) or
  E.g. 2: response = openai_client.chat.completions.create(model="gpt-4o", messages=messages).
- Developed and orchestrated sophisticated autonomous AI agents using frameworks like LangGraph, AutoGen, and Semantic Kernel, and leveraging inter-agent communication protocols and tools like n8n to integrate with external APIs, driving operational efficiency and intelligent decision-making.
  E.g, response = await kernel.invoke_async("email_plugin.summarize", text=email_body).
- Architected and deployed enterprise-grade Generative AI solutions using Amazon Bedrock, leveraging a suite of pre-trained foundation models including Anthropic's Claude 4 for its long context window and Amazon Titan. Applied prompt engineering and LLM-based RAG to build a context-aware customer support chatbot, reducing response time by 30%.
- Utilized cutting-edge Graph-R1 and Reinforcement Learning to build a robust Agentic Graph RAG framework, successfully reducing hallucination rates by configuring security-level parameters.
- Developed a comprehensive observability framework for data platforms, leveraging insights from CloudWatch Logs to predict ETL failures in Airflow and Glue jobs and implementing automated remediation patterns to reduce incident response time by over 40%. E.g., boto3.client('lambda').invoke(FunctionName='AutomatedRemediationFunction', InvocationType='Event', Payload='{"failure_type": "ETL_Error", "job_id": "job_123"}')
- **AI-Powered Automation & Insights**
- Engineered intelligent RPA workflows powered by NLP for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Integrated MongoDB Atlas as a vector database for RAG (Retrieval-Augmented Generation), enabling real-time and context-aware enterprise insights.
- **Machine Learning & Model Development**
- Developed, trained, and deployed a suite of machine learning and deep learning models on Amazon SageMaker using PyTorch, TensorFlow, and Hugging Face Transformers, optimizing data pipelines and automating model retraining to improve predictive accuracy and deliver NLP solutions.
- **DevOps, Cloud Infrastructure & Backend Development**
- Implemented cloud-native architectures with AWS ECS/EKS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms streamlined data ingestion within an AWS Redshift-powered lakehouse.
- Pioneered the use of LLMs and agentic workflows to automate and refactor code within scalable CI/CD pipelines, reducing deployment time by 25%. Engineered and deployed mission-critical Python applications and GenAI platforms on a multi-cloud architecture (AWS, Azure, GCP), leveraging IaC, MCP+RAG, and n8n orchestration.
  - E.g., llm_output = gemini.generate_text(f"Refactor this code to optimize performance: {code_snippet}")
  - E.g., ansible-playbook provision_rhel_app_servers.yml
  - E.g., requests.post(f"{api_url}/api/v1/deployments", json={'app':'genai-service', 'version':'v1.0.0'}).
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs).

Zhi Ming ZHANG - Australian Citizen
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- Pioneered the use of Large Language Models (LLMs) as optimizers within CI/CD pipelines to dynamically generate, test, and refactor code, leading to a 25% reduction in deployment time and a significant improvement in code quality across the team. E.g., llm_output = bedrock_client.invoke_model(modelId='anthropic.claude-v4', or 'amazon.titan-text-express-v1' body={ "prompt": "Refactor this code to optimize performance: {code_snippet}"
- **Technical Leadership & Innovation**
- Led end-to-end GenAI solution development from secure audio ingestion, automated transcription, LLM orchestration, RAG pipeline integration, to intelligent summarization—ensuring seamless data flow across multi-cloud.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Sr. AI/ML Engineer/Automation Engineer/Sr. DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- AI/ML-Powered Automation & Insights
- Developed self-learning Multi-agent systems (MAS), a subfield of distributed AI in Python to automate and optimize algorithmic strategies, enabling models to autonomously adapt to dynamic financial market patterns.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- High-Performance & Polyglot Software Engineering (Rust, Go, Java, Python)
- Designed and deployed a containerized financial fraud detection service, utilizing Rust for the ultra-low-latency transaction analysis and state management kernel, orchestrated via a scalable Python () layer, resulting in a 75% reduction in false positives compared to legacy systems.
  E.g., rust_core_response = requests.post(F'http://rust-svc/v1/fraud_check', json={'tx_id': id, 'data': payload}).json()
- Built real-time financial data ingestion and reconciliation pipelines using lightweight Go microservices for high-throughput messaging, feeding clean data directly into feature stores developed in Python for -driven anomaly detection models.
  E.g., go_consumer.read_msg().publish(python_feature_store, event.ReconciledData)
- **DevOps, Cloud Infrastructure & Backend Development**
- Pioneered the use of as optimizers to dynamically generate, test, and refactor code within scalable pipelines, achieving a 25% reduction in deployment time. Engineered and deployed mission-critical platforms on a multi-cloud architecture (AWS, GCP,Azure), leveraging and orchestration. E.g., ansible-playbook provision_rhel_app_servers.yml
- Implemented cloud-native architectures with AWS for containerized microservices and REST , ensuring efficient model integration and data ingestion within an -powered lakehouse.
- Developed a comprehensive observability framework for data platforms, leveraging insights from CloudWatch Logs to predict failures and implementing automated remediation patterns to reduce incident response time by over 40%.
  E.g., boto3.client('lambda').invoke(FunctionName='AutomatedRemediationFunction', InvocationType='Event', Payload='{"failure_type": "ETL_Error"}')
- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing and natural language processing. E.g., model.predict(vectorizer.transform(["RBA interest rate hike triggers AUD volatility"]))
- Led end-to-end machine learning solution development from secure audio ingestion, automated transcription, to intelligent summarization—ensuring seamless data flow across multi-cloud.Integrated speech-to-text APIs for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Developed computer vision models using OpenCV & Scikit-image to integrate image recognition for automated validation of property documents & anomaly detection, streamlining regulatory workflows & enhancing decision-making accuracy.
- **Technical Foundations & Modeling**
- Built robust predictive models and secure machine learning APIs using diverse algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive next best action (NBA) recommendations and pattern discovery.
- Enhanced algorithmic trading strategies and risk mitigation by discovering new market principles beyond human intuition.Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for machine learning services with core code such as requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'}).
- **DevOps & Cloud Infrastructure**
- Automated the creation of secure, scalable IaC CI/CD pipelines and isolated machine learning platforms, enabling robust deployment of Python (Django, Flask) and other server-side applications across AWS, Azure, and GCP.
- Implemented cloud-native architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within a data lakehouse.
- Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, and RHEL9 environments.
  E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- Leveraged PagerDuty as a central incident management system, effectively aggregating and transforming alerts from diverse monitoring and observability platforms (e.g., Airflow and Glue ETL failures, Datadog, Prometheus, AWS services) into actionable incidents.

- Integrated PagerDuty with Amazon DevOps Guru to leverage its AIOps/ML-powered operational insights, translating them into actionable incidents, triggering a PagerDuty incident. Runbook Automation via Bamboo: Reduce support costs by 50% by eliminating toil with PagerDuty. E.g., aws cloudwatch put-metric-data --metric-name CriticalApplicationError --namespace MyAIOpsApplication --value 1 --unit Count --dimensions Environment=Production,Service=AuthService.
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations. E.g. 1: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
  E.g. 2: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. ML Engineer/Automation Engineer /Integration Solution Architect /ML/DL/Mulesoft Developer
Responsibilities:

- **Automated Systems and Machine Learning**
- Engineered self-learning Multi-agent systems (MAS) in Python to automate the management of student enrollment and financial data, enabling models to autonomously adapt to complex patterns for enhanced system performance.
- Engineered intelligent rule-based Robotic Process Automation (RPA) workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed computer vision models (using OpenCV and Scikit-image) to integrate image recognition for the automated analysis of visual data and anomaly detection, streamlining data processing and enhancing decision-making accuracy.
- Led end-to-end machine learning project development, from data collection and automated transcription to statistical analysis and intelligent summarization, ensuring efficient data flow for research and academic projects.
- Developed scalable Python applications, leveraging core frameworks like scikit-learn, NLTK, Theano, and Caffe for data preprocessing, modeling, and natural language processing. Utilized third-party speech-to-text services for automatic transcription and sentiment analysis, integrating results into custom-built data analysis pipelines.
- **Machine Learning and Data Insights**
- Built robust predictive models and secure APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning). Applied Scikit-learn and Keras for projects focused on pattern discovery and predictive analytics.
- Utilized GPU-accelerated systems to develop and optimize machine learning models for research projects. Developed and tested algorithmic strategies by applying emerging principles in statistical analysis. Integrated databases like PostgreSQL or MongoDB for managing and analyzing data in projects, enabling real-time and context-aware insights.
- **Systems Engineering and Automation**
- Implemented cloud-based architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion.
- Automated the creation of secure and scalable infrastructure for machine learning experimentation using Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (e.g., Jenkins) for robust deployment across cloud platforms like AWS, Azure, and GCP.
- Used AWS infrastructure (EC2, S3, and RDS) to design, build, and maintain scalable systems for hosting ML models.
- Designed and developed scalable backend APIs and integrations using Python frameworks (Django, Flask), optimizing data pipelines for efficient machine learning model training and enabling near real-time predictions on datasets.
- Developed an intelligent web scraper (Python, requests, BeautifulSoup, pandas) with a simple UI built using Django or Flask to demonstrate robust data acquisition and processing capabilities.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: ML Engineer /Automation Engineer/Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- **Application Development and Machine Learning**
- Developed self-learning multi-agent systems in Python to automate and optimize the processing of injury claims, enabling autonomous adaptation to claims data and improving overall system efficiency.
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.
- Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. Proficient in Python and Java, with capabilities to design, develop, and optimize ML applications and deliver actionable insights.
- Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.

- Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Extensive expertise in programming languages such as Java, Python, C#, and PowerShell scripting languages.Data Engineering & Architecture
  Architected and implemented data solutions on AWS, focusing on traditional data warehousing and ETL processes.
- Utilized Amazon EMR for batch processing of large data volumes, and AWS Redshift for near-real-time data analysis.
- Employed Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created and implemented mapping documents that defined how source data fields corresponded to target data fields, including writing code to map the data according to defined rules.
- DevOps & Automation
- Automated deployment of scalable machine learning applications and services, leveraging AWS services like EC2 to ensure high availability.Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps/Data Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.
- Automated build and deployment processes using Jenkins and shell scripting, managing server configurations with Puppet to ensure reliable delivery of machine learning models for research projects.
- Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java.
- Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- **Security and Compliance**: Implemented security best practices and regularly patched systems and applications.
- **Agile and Data Expertise**: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Sr. Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Sr. Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.

- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Sr. Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI API , ChatGPT-4o, Multi-GPTs, Auto-Agents, Amazon Bedrock (Claude/Titan), SageMaker & AWS Kiro; Azure OpenAI Service, Azure ML, Azure AI Studio, Copilot, Azure AI Foundry, Google Vertex AI, Gemini API (formerly MakerSuite) , Agent-2-Agent
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
- **Workflow & Automation Platforms**: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve